THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 10, 2019 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) Describe the concept of "build, own, operate and transfer", and discuss whether this procurement method would suit the construction of a large scale wind farm facility from the perspective of the government and citizens. [10 marks]

   b) If the government has sufficient financial resources to fund the wind farm project, discuss whether it would be better to use the traditional procurement method or design and build by considering the pros and cons as well as the characteristics of these two procurement methods. [15 marks]

2. a) Calculate the maximum values of the construction site based on the following parameters:

   Gross floor area = 9,800m^2
   Floor area for lift lobby and corridor = 1,800m^2
   Floor area for lift machine room and water tank = 200m^2
   Construction cost = \$8,000/m^2
   Design and tendering period = 6 months
   Construction period = 16 months
   Expected usage = rental at \$200/m^2 per month
   Letting void = 2 months
   Interest rate for short-term finance = 12% per annum
   Initial yield = 5% [20 marks]

   b) As the construction cost would affect the accuracy of calculation in (a), describe the factors which could lead to a change in construction cost. [5 marks]

3. a) According to the American Society of Civil Engineers, *"civil engineers serve competently, collaboratively, and ethically as master planners, designers, constructors, and operators of society's economic and social engine"*. In the light of this statement, discuss the roles of civil engineers at different project stages to safeguard the best interests of the client and the society. [18 marks]

   b) Discuss whether it is ethical for contractors to *"form ring"* or submit a *"cover price"* and how should consultant civil engineers deal with these situations competently? [7 marks]

4. a) Describe the characteristics of outline programme, master programme, and sequence / repetitive programme, and employ suitable examples to explain their usage. [18 marks]

   b) Briefly discuss how to monitor and control the progress of a construction project to prevent any deviation from the master programme. [7 marks]

5. a) Compute the unit rate for a 150mm thick in-situ reinforced concrete (1:2:4, 40mm aggregate) curved roof slab (with a radius of 20 metres) based on the basic cost data as shown below:

Material

Type	*Material price*
Cement	$900.00 / tonne
Sand	$130.00 / tonne
Aggregate	$110.00 / tonne

Labour

Type	*Wage*
Concrete vibration	$140.00 / hour
All other workers	$120.00 / hour

Location	*Labour Constants*	
	Man hour	*Gang hour*
Slab < 100mm thick	5.32	1.33
Slab 100-150mm thick	4.68	1.17
Slab 150-300mm thick	4.32	1.08

Plant

Type	*Hiring charge*
Mixer (output = 0.333 hour / m^3)	$180.00 / hour
Pump (output = 0.25 hour / m^3)	$240.00 / hour

Note: There have been some arguments that curved concrete slabs are more difficult to cast than typical concrete slabs and would require an extra 30% of effort to cast. [20 marks]

b) A contactor is considering adopting a pricing strategy when pricing the tender. Explain with the assistance of suitable examples what is *"front loading"* and *"rear loading"*, and how should the client deal with the front or rear loaded tenders. [5 marks]

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